

EXPERIENCE LAB INTERNSHIP PROGRAMME

HEALTHCONNECT CLINIC

WEEK 7 — ANALYTICS TESTING & REFINEMENT REPORT

Project Title

**Improving Patient Appointment Attendance and Healthcare Support Using
Data and AI**

Track: Data Analytics

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1. EXECUTIVE SUMMARY

Week 7 focused on testing, validating, and refining the analytical outputs developed during Week 6. The purpose of the testing was to ensure that important KPIs, analytical findings, dashboard calculations, and dashboard filters were producing reliable and consistent results.

The testing process included validation of core KPIs, testing of important analytical findings, dashboard filter testing, investigation of an inconsistent Previous No-Shows result, and refinement of the interpretation of small and potentially misleading segments.

The core dataset contains **5,000 appointments**. The validated number of no-show appointments is **2,423**, producing a validated overall no-show rate of **48.46%**.

The testing confirmed several important analytical patterns. Booking Lead Days and Previous No-Shows showed the strongest observed associations with no-show behaviour. Reminder status and reminder channel also showed differences in observed no-show rates. Age and gender showed comparatively smaller differences.

The testing also identified limitations that need to be considered when interpreting the findings. These include small segment sizes, missing values for some fields, the difference between association and causation, and a discrepancy between the validated 48.46% no-show rate and a 51% rate reported in an independent analysis.

Overall, Week 7 increased confidence in the HealthConnect analytical outputs by systematically testing important calculations and dashboard interactions while documenting remaining analytical limitations.

2. TESTING OBJECTIVES

The main objectives of Week 7 testing were to:

1. Validate important KPI calculations.

2. Confirm that dashboard values agree with the underlying dataset.
3. Test important dashboard filters and calculations.
4. Validate key findings identified during Week 6.
5. Investigate unexpected or inconsistent analytical results.
6. Check whether findings remain consistent across relevant segments.
7. Identify analytical weaknesses and possible sources of misleading interpretation.
8. Refine dashboard interpretation and presentation.
9. Reassess business recommendations based on validated findings.
10. Identify findings that can support the wider HealthConnect solution.
11. Document testing evidence, issues, actions, and retesting results.
12. Identify remaining limitations and recommendations for Week 8.

3. TESTING APPROACH

The Week 7 testing process followed a structured approach.

For each important analytical output, the following process was applied:

Test / Scenario → Expected Result → Actual Result → Pass/Fail → Issue Identified → Action Taken → Retest Result

Testing was conducted using the HealthConnect Power BI dashboard and underlying analytical outputs.

Evidence considered during testing included:

- Power BI dashboard visuals
- Dashboard slicers
- Appointment counts
- No-show rates
- KPI calculations
- Segment-level analysis
- Combined analytical tables
- Filter testing
- Comparison with independent analytical work
- Before-and-after interpretation
- Documented issues and refinements

The objective was not to recreate the entire Week 5 or Week 6 analysis. Instead, testing focused on the most important outputs that could affect business interpretation and future HealthConnect decisions.

4. WEEK 7 TESTING READINESS

4.1 Components Selected for Testing

The following Week 6 components were identified as requiring testing:

Component	Intended Purpose	Testing Objective	Connected Track
No-Show Rate by Age Group	Understand differences in attendance by age	Validate rates and interpretation	Data Analytics
No-Show Rate by Gender	Understand gender differences	Validate rates and segment sizes	Data Analytics
No-Show Rate by Previous No-Shows	Identify relationship between previous behaviour and future no-shows	Validate calculation and pattern	Data Analytics / Data Science
No-Show Rate by Booking Lead Days	Identify relationship between lead time and attendance	Validate grouped rates	Data Analytics / Data Science
Reminder Sent	Assess differences between reminder and non-reminder appointments	Validate filtered rates	Data Analytics
Reminder Channel	Compare observed no-show rates across channels	Validate channel calculations	Data Analytics
Previous No-Shows × Booking Lead Days	Examine combined risk patterns	Validate whether individual patterns remain visible together	Data Analytics / Data Science

Component	Intended Purpose	Testing Objective	Connected Track
Core KPIs	Confirm overall dashboard accuracy	Validate calculations against the dataset	Data Analytics

5. KPI VALIDATION

The core KPIs were validated against the HealthConnect dataset containing 5,000 appointments.

KPI	Validated Value	Result
Total Appointments	5,000	PASS
No-Show Appointments	2,423	PASS
No-Show Rate	48.46%	PASS
Cancellation Rate	5.26%	PASS
Attendance Rate	46.28%	PASS
Non-Attendance Rate	53.72%	PASS

The no-show rate was validated using:

$$2,423 \div 5,000 = 48.46\%$$

The non-attendance rate was also validated:

$$48.46\% + 5.26\% = 53.72\%$$

The attendance and non-attendance rates also sum to 100%.

Therefore, the core KPI calculations were considered validated.

6. ANALYTICAL VALIDATION RESULTS

6.1 Previous No-Shows

The Week 6 analysis indicated that patients with more previous no-shows generally had higher no-show rates.

The Week 7 test reproduced the following results:

Previous No-Shows	Appointments	No-Show Rate
0	2,921	44%
1	1,548	53%
2	438	59%
3	78	68%
4	12	67%
5	3	100%

Testing Result

PASS — with refinement

The overall pattern from 0 to 3 previous no-shows was reproduced.

The groups with four and five previous no-shows are very small. The four-no-show group contains only 12 appointments, while the five-no-show group contains only 3 appointments.

Therefore, these percentages should not be treated as stable estimates of the wider patient population.

Discrepancy Investigation

During earlier Week 7 work, a value of 57% had been recorded for the group with four previous no-shows.

The result was retested in Power BI.

The current dashboard returned:

- **12 appointments**
- **0.67 = 67% no-show rate**

The Week 6 value of 67% was therefore reproduced.

The earlier 57% result was not reproduced and was treated as an inconsistent intermediate result rather than a corrected value.

The validated dashboard value was retained as **67%**.

6.2 Booking Lead Days

Booking Lead Days was tested to determine whether the observed increase in no-show rates across longer booking periods could be reproduced.

Booking Lead Days	Appointments	No-Show Rate
0–9	817	29%
10–19	865	37%
20–29	810	44%
30–39	843	51%
40–49	826	62%
50–59	745	68%
60+	94	68%

Testing Result

PASS — validated with refinement

The overall upward pattern was confirmed.

The no-show rate increased from 29% for appointments booked 0–9 days in advance to 68% for appointments booked 50–59 days in advance.

The 60+ group also had a 68% no-show rate, meaning the analysis should not claim that the rate continues increasing beyond 60 days.

The 60+ group also contains only 94 appointments and should therefore be interpreted with some caution.

6.3 Reminder Sent

The Reminder Sent analysis was tested to confirm the observed difference between appointments where a reminder was recorded and appointments where no reminder was recorded.

Reminder Sent	Appointments	No-Show Rate
No	1,366	51%
Yes	3,644	47%

Testing Result

PASS — validated

The dashboard reproduced:

- No reminder: 1,366 appointments and 51% no-show rate.
- Reminder sent: 3,644 appointments and 47% no-show rate.

The observed difference is approximately four percentage points.

However, this is an observational finding and does not establish that reminders caused the lower no-show rate.

6.4 Reminder Channel

The Reminder Channel analysis was tested using the dashboard filter.

Reminder Channel	Appointments	No-Show Rate
None	1,366	51%
WhatsApp	1,101	50%
Email	533	48%
SMS	2,000	46%

Testing Result

PASS — validated

The SMS filter returned:

- **2,000 appointments**
- **0.46 = 46% no-show rate**

This reproduced the expected result.

SMS had the lowest observed no-show rate among the channels in the dataset. However, this does not prove that SMS itself caused the lower rate because other factors may influence which patients receive each channel.

6.5 Age Group

The Age Group analysis was validated using appointment counts and no-show rates.

Age Group	Appointments	No-Show Rate
18–24	564	50%
25–34	783	51%
35–44	819	48%
45–54	793	48%
55–64	800	51%
65+	1,241	45%

Testing Result

PASS — validated

The Age Group filter was tested by selecting **65+**.

The connected gender visual returned:

- Female: 621 appointments, 44%
- Male: 592 appointments, 46%
- Prefer not to say: 28 appointments, 36%

The counts sum to:

$$621 + 592 + 28 = 1,241$$

This matches the total number of 65+ appointments.

The test also demonstrated that the 65+ group had the largest appointment volume but not the highest no-show rate.

This reinforces the importance of considering rates alongside raw counts.

6.6 Gender

Gender was tested to validate whether the Week 6 rates were reproduced.

Gender	Appointments	No-Show Rate
Female	2,488	48%
Male	2,404	49%
Prefer not to say	108	44%

Testing Result

PASS — validated

Female and male no-show rates differed by only one percentage point.

The "Prefer not to say" group was considerably smaller and should therefore be interpreted cautiously.

Gender does not show a large difference in this dataset and should primarily remain a descriptive variable rather than the main basis for operational intervention.

6.7 Previous No-Shows × Booking Lead Days

A combined analysis was tested to determine whether the two important individual patterns remained visible when analysed together.

Among patients with **zero previous no-shows**, the observed rates included:

- 0–9 days: **24%**
- 50–59 days: **62%**
- 60+ days: **65%**

This demonstrates that the booking lead-time pattern remained visible even among patients with no previous no-show history.

Some combined cells produced extreme values of 100%. However, several of these cells contained only 2–3 appointments.

Testing Result

PASS — with significant refinement

The combined analysis supported the broader patterns observed individually.

However, very small cells should not be treated as stable estimates.

The combined analysis therefore supports future modelling considerations but requires careful treatment of small samples.

7. DASHBOARD FILTER TESTING

Dashboard slicers were tested to ensure that filtering changed the connected visuals correctly and that clearing filters returned the dashboard to its overall state.

7.1 Age Group Filter

Test: Select Age Group = 65+

Expected: 1,241 appointments and corresponding filtered values.

Actual:

- Female: 621
- Male: 592
- Prefer not to say: 28
- Total: 1,241

Result: PASS

The filter correctly restricted the connected visual.

7.2 Booking Lead Days Filter**Test:** Select Booking Lead Days = 50–59.**Expected:**

- 745 appointments
- 68% no-show rate

Actual:

- 745 appointments
- $0.68 = 68\%$

Result: PASS

The filter reproduced the validated result.

7.3 Reminder Sent Filter

Test: Select Reminder Sent = No.

Expected:

- 1,366 appointments
- 51% no-show rate

Actual:

- 1,366 appointments
- $0.51 = 51\%$

Result: PASS

The filter and calculation worked correctly.

7.4 Reminder Channel Filter

Test: Select Reminder Channel = SMS.

Expected:

- 2,000 appointments
- 46% no-show rate

Actual:

- 2,000 appointments
- $0.46 = 46\%$

Result: PASS

The filter reproduced the validated result.

7.5 Clear-Filter Test

After testing individual filters, all filters were cleared.

Expected: Dashboard should return to the overall dataset.

Actual:

- Total appointments: **5,000**
- No-show rate displayed: **0.48**

The exact validated KPI is **48.46%**, while the dashboard displayed the value as a decimal.

Result: PASS

The dashboard returned to the overall dataset.

8. DASHBOARD REFINEMENT

The Week 7 process focused primarily on refining interpretation and validating dashboard behaviour rather than rebuilding the entire dashboard.

Area	Before Testing	After Testing	Reason
KPI calculations	Not systematically retested	Core KPIs validated	Confirm accuracy
Previous No-Shows	Small groups could be over-interpreted	Small-sample caution added	Avoid misleading conclusions
Booking Lead Days	Increasing pattern identified	60+ plateau documented	Improve interpretation
Reminder filters	Not formally tested	Reminder filters tested	Confirm dashboard interaction
Age Group	Raw counts could influence interpretation	Rates considered alongside counts	Improve analytical interpretation
Filter reset	Not formally evidenced	Clear-filter test completed	Confirm usability
Previous No-Shows = 4	Earlier note contained 57%	Dashboard-supported 67% retained	Resolve inconsistency

The main refinement was therefore the strengthening of the analytical interpretation and validation process.

9. VALIDATED KEY FINDINGS

9.1 Booking Lead Time

Testing confirmed a strong observed relationship between booking lead time and no-show rates.

The rate increased from **29% at 0–9 days** to **68% at 50–59 days**.

This indicates that long-lead appointments may require additional confirmation and monitoring.

The analysis does not establish that booking lead time itself causes no-shows.

9.2 Previous No-Show History

No-show rates generally increased as previous no-show history increased.

The rate increased from:

44% with zero previous no-shows

to:

68% with three previous no-shows.

The 4- and 5-no-show groups are too small to support strong general conclusions.

9.3 Reminder Status

Appointments with a recorded reminder had an observed no-show rate of **47%**, compared with **51%** where no reminder was recorded.

This supports continued monitoring of reminder processes.

9.4 Reminder Channel

SMS had the lowest observed no-show rate at **46%**, followed by Email at 48%, WhatsApp at 50%, and no recorded reminder at 51%.

Further evaluation would be required before concluding that one channel causes better attendance.

9.5 Age

The 65+ group had the largest number of appointments but a lower no-show rate than some other age groups.

This demonstrates why HealthConnect should consider both **volume and rate** when interpreting attendance problems.

9.6 Gender

Female and male no-show rates were very similar:

- Female: 48%
- Male: 49%

Gender therefore showed a relatively small difference in this dataset.

10. UPDATED BUSINESS INSIGHTS

The Week 7 testing strengthened the interpretation of the Week 6 analysis.

The strongest observed analytical signals were associated with:

1. Booking Lead Days
2. Previous No-Shows
3. Reminder Status
4. Reminder Channel

Age and gender showed smaller differences.

The combined analysis also showed that booking lead time remained associated with no-show rates even among patients with zero previous no-shows.

These findings provide a stronger basis for HealthConnect to consider targeted attendance-support processes.

11. UPDATED RECOMMENDATIONS

11.1 Additional Confirmation for Long-Lead Appointments

Appointments booked 50 or more days in advance showed an observed no-show rate of approximately 68%.

HealthConnect could introduce additional confirmation steps for long-lead appointments.

Possible actions include:

- Initial confirmation after booking
- Reminder closer to the appointment date
- Additional confirmation request for particularly long-lead appointments

11.2 Targeted Follow-Up Based on Previous No-Shows

Patients with previous no-show history showed higher observed no-show rates.

HealthConnect could use previous no-show history as one factor in identifying appointments that may benefit from additional confirmation.

This should be supportive rather than punitive.

11.3 Continue Reminder Processes

The reminder-sent group had a lower observed no-show rate than the group without a recorded reminder.

HealthConnect should continue the reminder system and monitor its performance over time.

11.4 Evaluate Reminder Channels

SMS had the lowest observed no-show rate among the tested channels.

HealthConnect could investigate SMS further while avoiding the assumption that the channel itself caused the difference.

11.5 Develop a Risk-Based Analytical Approach

Potential variables for future predictive analysis include:

- Previous No-Shows
- Booking Lead Days
- Reminder Sent
- Reminder Channel
- Age Group
- Gender
- Appointment Type

These variables can be investigated in future Data Science work.

11.6 Use Rates Alongside Counts

Dashboard analysis should continue showing appointment volume together with no-show rates.

This reduces the risk of targeting groups simply because they have a large number of appointments.

11.7 Continue KPI Validation

HealthConnect should maintain a regular process for validating:

- Total appointments
- No-show rate
- Attendance rate
- Cancellation rate
- Reminder performance
- Segment sizes
- Dashboard filter behaviour

12. CROSS-TRACK CONTRIBUTION

12.1 Collaboration

The primary Week 7 collaboration was between **Data Analytics and Data Analytics**, through comparison with a fellow Data Analytics student's independent analysis.

The purpose of the comparison was to strengthen analytical validation and identify differences that required reconciliation.

12.2 Findings Shared

The following findings were shared and compared:

- Overall no-show rate
- Booking Lead Days
- Previous No-Shows
- Reminder Sent
- Reminder Channel
- Age Group
- Gender
- Combined segmentation
- KPI calculations
- Dashboard testing

12.3 Difference Identified

The independent analysis reported an overall no-show rate of approximately **51%**, while the validated dashboard calculation produced **48.46%**.

Rather than replacing one value with the other without evidence, the difference was documented as a reconciliation issue.

The validated 48.46% value was retained for this report because it was directly validated using the 2,423 no-show appointments out of 5,000 total appointments.

12.4 Contribution to the Wider HealthConnect Solution

The validated findings can support other HealthConnect workstreams.

Data Science

Potential candidate variables for future predictive modelling include:

- Previous No-Shows
- Booking Lead Days
- Reminder Sent
- Reminder Channel
- Age Group
- Gender
- Appointment Type

Operations

The findings can support:

- Additional confirmation workflows
- Targeted follow-up
- Reminder scheduling
- Attendance monitoring
- Appointment management

AI / Decision Support

The validated analytical findings can provide evidence for future AI-assisted decision-support tools.

Any future AI recommendations should be based on validated data and should not treat observed associations as proof of causation.

13. TESTING CHANGES AND ISSUE LOG

Issue	Impact	Action Taken	Status
Small Previous No-Shows groups	Extreme percentages may be unstable	Added sample-size caution	Resolved through refinement
60+ Booking Lead Days	Could be incorrectly interpreted as continued increase	Documented the plateau at 68%	Resolved
Raw counts vs rates	Could lead to incorrect targeting	Emphasized rates alongside counts	Resolved
Previous No-Shows = 4 discrepancy	Earlier 57% value conflicted with dashboard 67%	Retested and retained 67%	Resolved for dashboard
Decimal KPI display	0.48 is less intuitive than 48.46%	Confirmed underlying KPI and recommended percentage formatting	Refinement recommended
Independent KPI discrepancy	51% vs 48.46%	Flagged for reconciliation	Open

14. LIMITATIONS

14.1 Missing Data

The dataset contains:

- **90 blank Distance to Clinic values**
- **60 blank Waiting Time values**

These missing values were not replaced with zero because zero would incorrectly imply no distance or no waiting time.

14.2 Small Segment Sizes

Some analytical groups are very small.

For example:

- Previous No-Shows = 4 → 12 appointments
- Previous No-Shows = 5 → 3 appointments
- Some combined segmentation cells → 2–3 appointments

Extreme percentages in these groups should therefore be interpreted cautiously.

14.3 Association Does Not Mean Causation

The analysis identifies relationships between variables and appointment outcomes.

It does not prove that:

- Long booking lead time causes no-shows
- Previous no-shows cause future no-shows
- Reminders cause lower no-show rates
- SMS causes better attendance

Further research or controlled evaluation would be required to establish causal effects.

14.4 Reminder Confounding

Patients receiving reminders may differ from patients who do not receive reminders in other ways.

Therefore, the difference between 47% and 51% should not be interpreted as proof that reminders directly caused improved attendance.

14.5 Historical Dataset

The findings are based on the available historical HealthConnect dataset.

Future appointment behaviour may differ, so continued monitoring is required.

14.6 KPI Reconciliation

An unresolved difference exists between:

Validated dashboard KPI: 48.46%

and

Independent analysis: approximately 51%

The difference should be investigated by comparing:

- Dataset version
- Outcome definitions
- Filters
- Formulas
- Records included
- Missing-value treatment

14.7 Dashboard Display Precision

The Week 7 dashboard displayed the no-show rate as **0.48** rather than **48.46%**.

The underlying KPI was validated as 48.46%.

For improved usability, the dashboard should format the measure as a percentage.

15. REMAINING ANALYTICAL ISSUES

The following issues remain relevant after Week 7 testing:

1. Reconciliation of the 48.46% and 51% overall no-show rates.
2. Further investigation of small Previous No-Shows groups.
3. Further evaluation of reminder effectiveness.
4. Continued monitoring of Booking Lead Days patterns.
5. Investigation of missing Distance and Waiting Time values.

6. Continued validation of dashboard KPIs.
7. Improved percentage formatting on the dashboard.
8. Further evaluation before using the findings for predictive or automated decision-making.

16. WEEK 8 RECOMMENDATIONS

Week 8 should build on the validated Week 7 findings.

16.1 Reconcile the KPI Difference

Investigate the difference between the 48.46% and 51% no-show rates before combining the two analyses.

16.2 Explore Predictive Analytics

Investigate whether a predictive model can identify appointments with higher no-show risk.

Candidate variables include:

- Previous No-Shows
- Booking Lead Days
- Reminder Sent
- Reminder Channel
- Age Group
- Gender
- Appointment Type

16.3 Evaluate Reminder Effectiveness

Investigate reminder timing, frequency, and channel effectiveness.

16.4 Continue Dashboard Monitoring

Continue monitoring validated KPIs and key segments over time.

16.5 Improve Dashboard Presentation

Improve percentage formatting and clearly distinguish between:

- Appointment volume
- No-show rate
- Attendance rate
- Cancellation rate

16.6 Monitor Small Segments

Use sample-size warnings or minimum-count thresholds before making operational decisions based on very small groups.

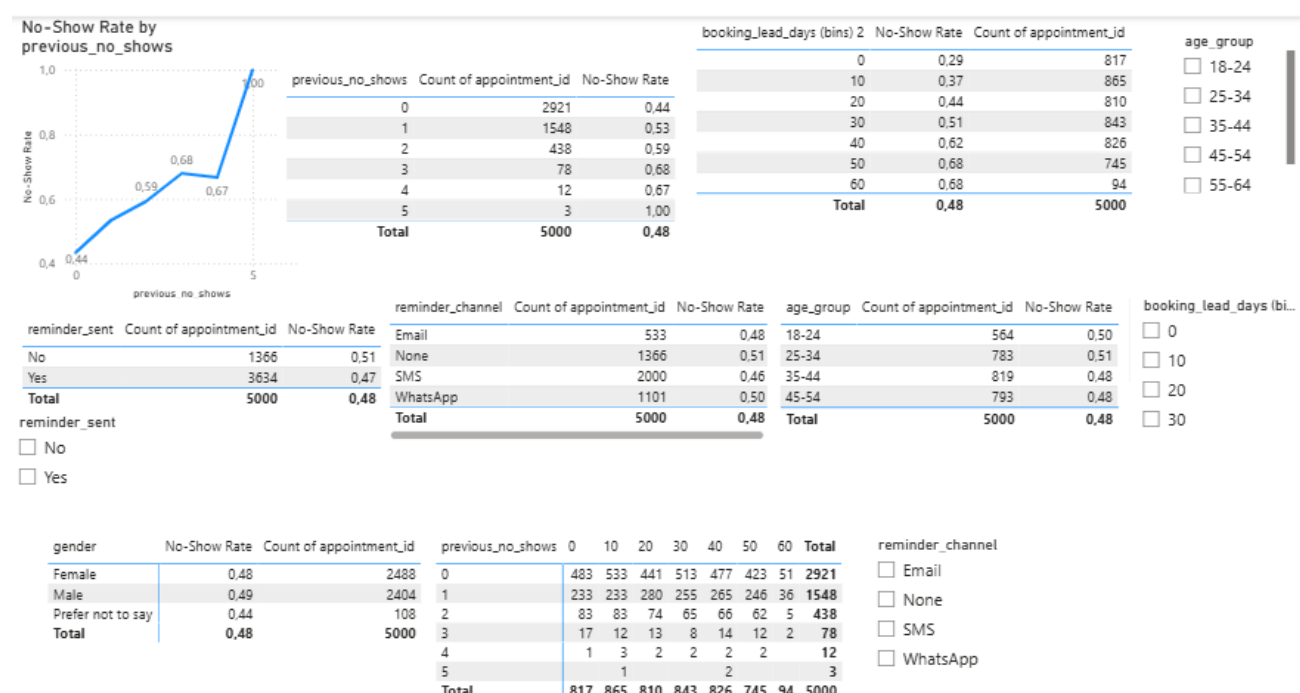
16.7 Support Wider HealthConnect Tracks

Provide validated analytical findings and candidate variables to relevant Data Science, Operations, and AI workstreams.

17. TESTING EVIDENCE

A consolidated Power BI screenshot was used to document the Week 7 analytical testing and dashboard refinement process.

Figure 1: Consolidated Week 7 analytical testing and dashboard validation evidence.



The evidence includes the following validated analytical outputs:

- No-Show Rate by Previous No-Shows
- No-Show Rate by Booking Lead Days
- No-Show Rate by Reminder Sent
- No-Show Rate by Reminder Channel
- No-Show Rate by Age Group
- No-Show Rate by Gender
- Previous No-Shows × Booking Lead Days analysis
- Appointment counts alongside no-show rates
- Overall dataset totals
- Dashboard filtering controls

The dashboard evidence shows the underlying appointment counts and corresponding no-show rates used during Week 7 testing. The analytical outputs reproduced the major Week 6 patterns, while the testing process also identified small sample sizes, the Booking Lead Days 60+ plateau, and the importance of interpreting rates alongside appointment counts.

The dashboard contains slicers for Age Group, Booking Lead Days, Reminder Sent, and Reminder Channel. These filters were separately tested during Week 7 to confirm that the dashboard responded correctly to selected segments and returned to the overall dataset after filters were cleared.

The consolidated screenshot therefore provides visual evidence of the analytical outputs used for validation and refinement, while the documented testing results provide the detailed record of the individual test scenarios, expected results, actual results, and conclusions.

18. FINAL WEEK 7 CONCLUSION

Week 7 successfully tested and validated important HealthConnect analytical outputs and dashboard interactions.

The core KPI validation confirmed **5,000 total appointments, 2,423 no-show appointments, and a 48.46% overall no-show rate.**

The analytical testing confirmed important patterns involving Booking Lead Days, Previous No-Shows, Reminder Sent, and Reminder Channel. The testing also demonstrated that raw appointment counts should not be interpreted independently of no-show rates.

Dashboard filter testing successfully reproduced expected results for Age Group, Booking Lead Days, Reminder Sent, and Reminder Channel. Clearing the filters also returned the dashboard to the overall dataset.

The investigation of the Previous No-Shows = 4 discrepancy demonstrated the importance of retesting unexpected results. The current dashboard reproduced **67% based on 12 appointments**, so this value was retained rather than using the previously recorded 57%.

The main remaining analytical issue is the discrepancy between the independently reported **51%** overall no-show rate and the validated **48.46%** dashboard KPI. This should be reconciled before the results are combined.

Overall, the Week 7 testing process improved confidence in the HealthConnect analytical outputs while identifying important limitations around sample size, missing data, causality, and KPI reconciliation.

The validated findings provide a stronger analytical foundation for future HealthConnect work, including appointment management, reminder strategies, predictive analytics, and AI-supported decision-making.